

QWeave Three Week Team Execution Plan

Four-person work division, integration rhythm, and submission gates

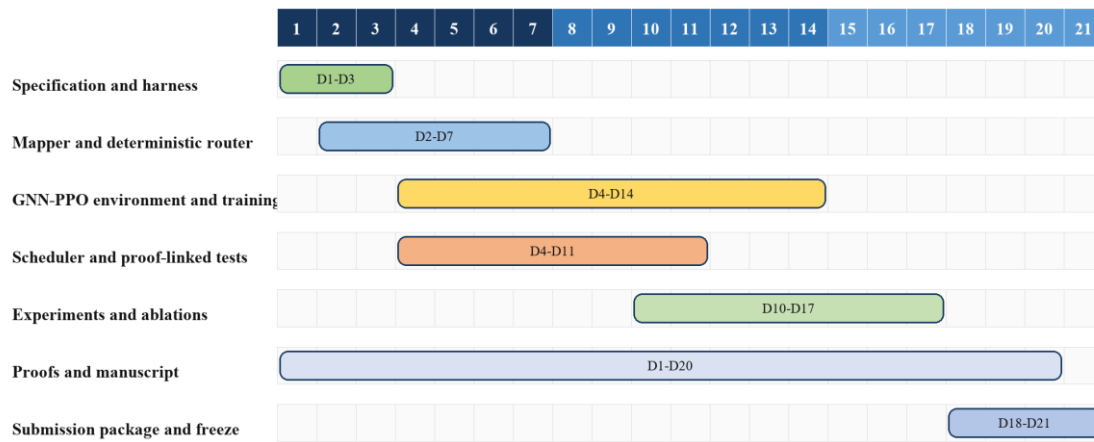
Team: Shrivardhini N | Atharva Sheersh Pandey | Diptesh Das | Haridasu Sreedhar

Guide: Prof. Bhuvaneswari M | Planning horizon: 21 days | Version: 1.0

Operating Principle

The team should work in parallel but integrate daily. Each workstream has one directly responsible owner and one reviewer. Proofs are not a separate writing exercise at the end: every formal invariant must be paired with an implementation assertion and test during development. The allocations below are a balanced starting point and may be swapped if skills differ.

QWeave 21-Day Critical Path



Hard freeze: Day 17 results | Day 19 paper | Day 21 repository and submission bundle

Figure 1. Twenty-one-day critical path. Workstreams overlap, but result and manuscript freezes are fixed.

1 Role Allocation

Person	Primary ownership	Secondary ownership	Final accountable outputs
Atharva Sheersh Pandey	Integration lead; Qiskit IR, target adapter, experiment harness	Repository, CI, final artifact build	End-to-end compiler command; immutable experiment records; release tag
Shrivardhini N	Mathematical lead; formal model, proofs, equivalence strategy	Paper methods and limitations	LaTeX theorem section; proof-to-test matrix; correctness review
Diptesh Das	Deterministic optimisation lead; initial mapping, fallback router, baselines	Small-instance CP-SAT oracle	Mapper and fallback modules; SABRE/oracle comparisons; ablation outputs
Haridasu Sreedhar	Learning lead; Gymnasium environment, GNN, masked PPO	Training diagnostics and topology generalisation	Policy code and checkpoints; training logs; learning ablations

Guide checkpoint. Prof. Bhuvaneswari M should review the frozen research question on Day 2, the proof boundary and preliminary pipeline on Day 8, the result interpretation on Day 17, and the final manuscript on Day 20.

2 Review Pairing and Bus-Factor Control

Workstream	Owner	Reviewer	Required handoff artifact
Compiler integration	Atharva	Diptesh	Interface tests and sample routed circuit
Formal proofs	Shrivardhini	Atharva	Assumption list, theorem source, linked assertions
Mapper/fallback/baselines	Diptesh	Shrivardhini	Objective trace, termination tests, baseline parity
GNN-PPO	Haridasu	Atharva	Environment checker, masked-policy tests, training dashboard
Scheduler	Shrivardhini	Diptesh	Feasibility tests, lower-bound calculations, makespan report
Statistics and figures	Atharva	Haridasu	Frozen analysis script and regenerated figures
Manuscript	Shrivardhini	All authors	Compiled PDF, response checklist, contribution statement

3 Daily Plan

Day	Focus	Parallel assignments	Exit evidence
1	Kick-off and scope freeze	Atharva: repo/env spike. Shrivardhini: formal objects. Diptesh: baselines and mapper spec. Haridasu: RL state/action draft.	One-page research contract; repository skeleton; initial bibliography.
2	Interfaces and datasets	Agree dataclasses and target adapter. Select circuits/topologies and create train/validation/test manifest.	Interface document; immutable split manifest; guide review.
3	Baseline gate	Run Qiskit SABRE and basic routing on smoke circuits. Establish metrics and validation skeleton.	Baseline CSV/JSON; reproducible seed; Gate A passed.
4	Core implementations	Mapper and scheduler coding; Gymnasium environment scaffold; theorem assumptions and invariants.	Unit tests for graphs, mapping, ready set, resources.
5	Deterministic routing	Implement fallback shortest-path router and final-layout bookkeeping; policy feature builder.	Legal routed outputs on paths/grids; mapping trace.
6	End-to-end deterministic path	Integrate mapper, fallback, scheduler, validation; implement GNN actor-critic forward pass.	One command compiles smoke suite; PyG tensors validated.
7	Core gate	Stress deterministic pipeline and exact oracle on tiny cases; fix interfaces before learning integration.	Gate B passed; no open interface redesign.
8	Masked policy integration	Connect GNN scores to edge actions and EXECUTE; prove/verify zero probability for illegal actions.	Environment checker passes; action-mask property tests.
9	PPO rollout	Implement trajectory buffer, GAE, clipped loss, entropy and value losses; start tiny curriculum.	Short training run improves shaped return without invalid actions.
10	Scheduler feedback	Feed realised or incremental depth signals into routing reward; monitor fallback and cycles.	Reward decomposition logs; scheduler lower-bound report.
11	Learning gate	Run end-to-end learned routing on validation circuits; profile bottlenecks and memory.	Gate C learning milestone; selected architecture fixed.
12	Curriculum and tuning	Limited hyperparameter search on validation set only; expand circuit width/topology diversity.	Candidate configs and deterministic comparison notebook/script.
13	Robustness	Topology holdout evaluation, failure analysis, watchdog tuning, proof assumption audit.	Generalisation table draft; failure taxonomy.
14	Training freeze	Choose one primary model and one backup. Freeze reward weights, architecture, seeds, and timeouts.	Signed configuration hash; test set still unopened.

Day	Focus	Parallel assignments	Exit evidence
15	Main experiments	Run all baselines and QWeave on test circuits. Validate every output and preserve failures.	Per-circuit raw result bundle, run manifests, logs.
16	Ablations and statistics	Run required ablations; calculate paired deltas, bootstrap intervals, tests, and effects.	Tables and main plots regenerated from scripts.
17	Results freeze	No method changes. Review anomalies, timeouts, and negative results; write exact claims.	Gate D passed; result table frozen and checksummed.
18	Paper assembly	Integrate methods, proofs, experiments, results, threats, data statement, and related work.	Complete first full manuscript; all figures final-size readable.
19	Manuscript freeze	Internal review for claim/evidence alignment, citations, theorem assumptions, and reproducibility.	Gate E draft; no new experiments unless correcting an error.
20	Guide and venue review	Apply guide feedback; format target template; anonymise if required; verify quartile/scope and costs.	Submission-ready manuscript, supplement, cover letter draft.
21	Release and submission bundle	Fresh-clone reproduction test; tag repository; archive release; final author approvals.	Release artifact, final PDFs/source, checklist, submission package.

4 Person-Level Deliverables

Atharva Sheersb Pandey

- Qiskit pass integration and CLI
- Target/circuit adapters and shared dataclasses
- Experiment runner, configuration hashing, CI, release
- Statistical aggregation and regenerable figures
- Final repository and artifact reproducibility test

Shrivardhini N

- Formal problem model and assumption register
- Mapping, legality, equivalence, scheduler, and termination proofs
- Depth-aware scheduler and feasibility checker
- Proof-linked property/equivalence tests
- Methods, theory, limitations, and final manuscript integration

Diptesh Das

- Weighted interaction graph and initial mapping
- Strict-improvement local search and trace
- Deterministic shortest-path fallback
- SABRE/basic baselines and small CP-SAT oracle
- Mapper/fallback/oracle ablations and runtime analysis

Haridasu Sreedhar

- Gymnasium environment and graph observations
- PyG message-passing actor-critic
- Masked edge-action distribution and PPO trainer
- Curriculum, checkpoints, diagnostics, and topology holdout
- No-GNN/no-shaping/no-learning ablations

5 Daily Integration Rhythm

- 09:00-09:15: stand-up. State yesterday's evidence, today's measurable exit, and blockers.
- 13:00: interface checkpoint. Merge contract changes before dependent work continues.
- 18:00: integration window. Owner demonstrates tests/results; reviewer approves or returns specific issues.
- 18:30: automated smoke run and artifact upload. Main must remain green overnight.
- Twice weekly: 30-minute claim audit. Each manuscript sentence labelled proof, measured result, interpretation, or future work.

5.1 Pull Request Rules

- One logical change per pull request; no direct commits to main after Day 3.
- Every algorithm change includes or updates tests and a short complexity/assumption note.
- Every experiment change updates a versioned config; never overwrite raw results.
- Proof-affecting code requires mathematical-lead review. Metric-affecting code requires integration-lead review.
- A reviewer reproduces one representative command before approval.

6 RACI for Critical Decisions

Decision	Responsible	Accountable	Consulted	Informed
Research question and scope	All	Guide + Atharva	All	All
Formal assumptions and theorem wording	Shrivardhini	Shrivardhini	Atharva, Diptesh	Haridasu
Compiler interfaces	Atharva	Atharva	Diptesh, Haridasu	Shrivardhini
Reward and model architecture	Haridasu	Haridasu	Shrivardhini, Atharva	Diptesh
Baseline and oracle settings	Diptesh	Diptesh	Atharva, Shrivardhini	Haridasu
Results freeze and paper claims	All	Guide + Shrivardhini	All	All
Venue and submission	Atharva, Shrivardhini	Guide	All	All

7 Risk and Contingency Plan

Risk trigger	Immediate response	Protected paper contribution
PPO fails to beat heuristic by Day 13	Stop broad tuning; use learned reranking on a small candidate set or present negative ablation honestly	Proof-backed safe compiler and rigorous evaluation remain
Training too slow	Reduce width/curriculum; parallelise environments; freeze deterministic core	Correctness, mapper, scheduler, oracle comparisons
Equivalence failures	Block experiments; isolate final-layout or direction/decomposition bug	No invalid result enters the paper
Exact solver times out	Restrict oracle to tiny instances and report optimality bounds/timeouts	Oracle remains a sanity reference
Insufficient novelty for Q1/Q2	Strengthen proof-to-implementation linkage and topology-holdout evidence; do not inflate claims	Submission may move to a better-fit venue after advisor review
One member unavailable	Reviewer assumes owner role using mandatory handoff artifacts	Daily integration prevents single-person knowledge silos

8 Freeze Checklists

8.1 Day 14 Training Freeze

- Primary and backup checkpoints selected
- Config, environment, split, and source commit hashes recorded
- No test data inspected
- Reward components and metric code unit-tested
- Fallback threshold fixed

8.2 Day 17 Results Freeze

- All outputs validated
- All failures/timeouts included
- Ablations complete
- Paired statistics regenerated
- Claims written with exact numbers and uncertainty
- Raw results immutable

8.3 Day 21 Submission Freeze

- Fresh-clone reproduction passes
- Paper source compiles without warnings affecting output
- Author names, affiliations, emails, order, and contributions approved
- Venue scope/quartile/cost rechecked
- Ethics, data, code, conflict, and AI-use statements complete
- Repository release archived and cited

9 Final Team Rule

If a feature does not strengthen the main claim, a proof obligation, a required baseline, or reproducibility, it is out of scope until after submission. Depth minimisation is the headline; safety, correctness, and transparent evidence are what make the headline publishable.